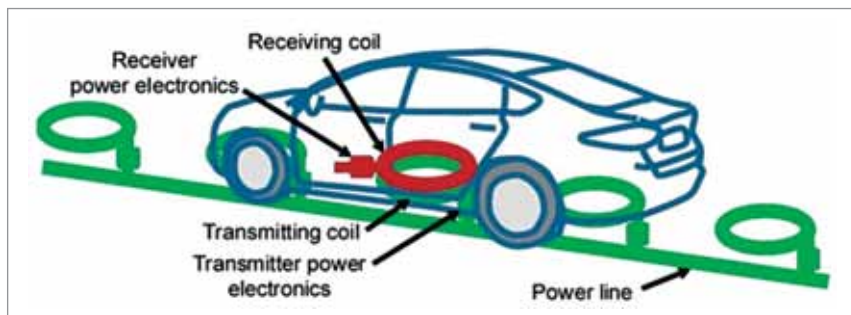
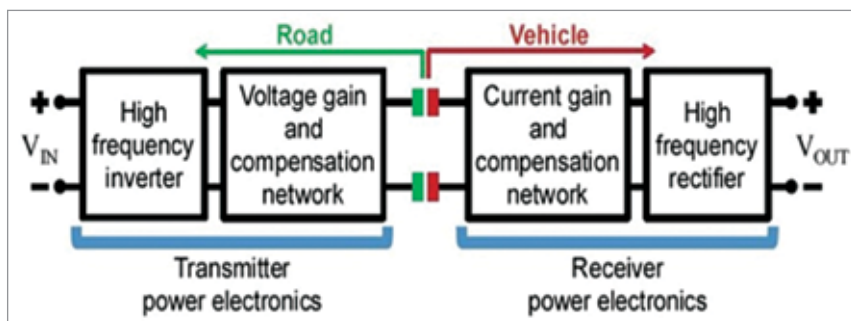


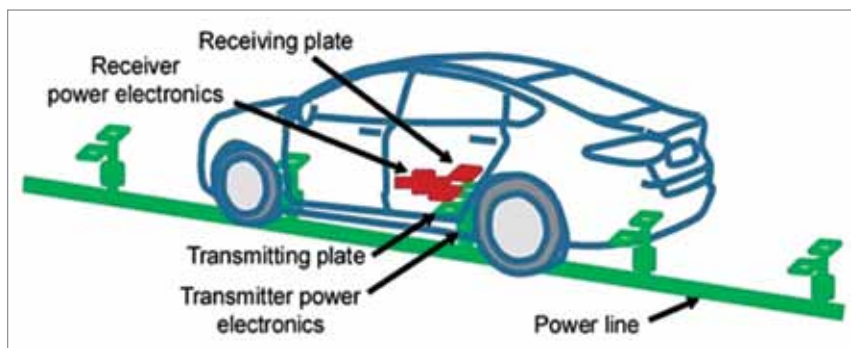
Block diagram of a typical inductive wireless power transfer system (Credit: www.ncbi.nlm.nih.gov)



The inductive system physically implemented in an EV (Credit: www.ncbi.nlm.nih.gov/books)



Block diagram of the capacitive wireless power transfer system (Credit: www.ncbi.nlm.nih.gov/books)



Physical implementation of a capacitive system in an EV (Credit: www.ncbi.nlm.nih.gov/books)

transmitter) is used to create a time-varying magnetic field. The magnetic field induces an alternating current in coils of wire (called receiver) placed in that magnetic field and electric energy gets transferred from the transmitter to the receiver through inductive coupling. Resonant charging (also called magnetic resonance) is used

in charging bowls in order to increase the induced current to transmit power efficiently across greater distances. Appropriate capacitances are added to the transmitter and receiver coils, so that they resonate at the frequency of the time-varying magnetic field.

To minimise the weight and decrease the skin effect resistance, a

movable transmission coil mounted on an elevating platform and receiver coils made of silver-plated copper or aluminium are used. The distance over which the power can travel also depends on the size of the coils used. The bigger the coil, or more the number of coils, the greater is the distance the charge can travel. In EV chargers, the coils used are of much larger diameter so that energy gets transferred over greater distance, in comparison with smartphone wireless charging.

Wireless charging systems

The wireless power transfer systems are of two types: inductive—which use magnetic field coupling between conducting coils, and capacitive—which use electric field coupling between conducting plates to transfer energy. The inductive system uses a transformer for electromagnetic coupling whereas a capacitive system uses a network with inductors and capacitors for the coupling. Electronic circuits, consisting of semiconductor rectifiers, high-frequency inverters, and gain and compensation networks are used in both the systems to convert the alternating current into direct current to charge the batteries.

Inductive systems are preferred for medium-range static charging systems, where the distance between the transmitters embedded in the roadway and the receiver fitted on the electric vehicle is comparable to the size of the couplers. Stationary chargers using inductive charging are already available in the market for use in cars. A few manufacturers are planning to introduce the built-in stationary inductive charging systems in their high-end cars.

In dynamic systems, where a truly short time is available when the vehicle is passing over a charging coil, high power transfer density capability is required to deliver the required energy to the vehicle during this time. For this reason, the dynamic power transfer systems have not been commercially viable, although a few experimental systems have been demonstrated.

Operating frequency

Designers prefer capacitive wireless charging systems, which operate at